Example 4.21

Prove that (i) $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}=\frac{\pi}{4}$ (ii) $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7}=\tan^{-1}\frac{31}{17}$

Solution

(i) We know that $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\frac{x+y}{1-xy}$, $xy<1$.

Thus, $\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{3}=\tan^{-1}\frac{\frac{1}{2}+\frac{1}{3}}{1-\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)}=\tan^{-1}(1)=\frac{\pi}{4}$.

(ii) We know that $2\tan^{-1}x=\tan^{-1}\frac{2x}{1-x^2}$, $-1<x<1$

So, $2\tan^{-1}\frac{1}{2}=\tan^{-1}\frac{2\left(\frac{1}{2}\right)}{1-\left(\frac{1}{2}\right)^2}=\tan^{-1}\left(\frac{4}{3}\right)$.

Hence, $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7}=\tan^{-1}\frac{4}{3}+\tan^{-1}\frac{1}{7}=\tan^{-1}\left(\frac{\frac{4}{3}+\frac{1}{7}}{1-\left(\frac{4}{3}\right)\left(\frac{1}{7}\right)}\right)=\tan^{-1}\left(\frac{31}{17}\right)$.

Example 4.22

If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$ and $0<x,y,z<1$, show that

$x^2+y^2+z^2+2xyz=1$

Solution

Let $\cos^{-1}x=\alpha$ and $\cos^{-1}y=\beta$. Then, $x=\cos\alpha$ and $y=\cos\beta$.

$\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$ gives $\alpha+\beta=\pi-\cos^{-1}z$. ... (1)

Now, $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta=xy-\sqrt{1-x^2}\sqrt{1-y^2}$.

From (1), we get $\cos\left(\pi-\cos^{-1}z\right)=xy-\sqrt{1-x^2}\sqrt{1-y^2}$

$-\cos\left(\cos^{-1}z\right)=xy-\sqrt{1-x^2}\sqrt{1-y^2}$.

So, $-z=xy-\sqrt{1-x^2}\sqrt{1-y^2}$, which gives $-xy-z=-\sqrt{1-x^2}\sqrt{1-y^2}$.

Squaring on both sides and simplifying, we get $x^2+y^2+z^2+2xyz=1$.

Example 4.23

If $a_1, a_2, a_3, \ldots a_n$ is an arithmetic progression with common difference d, prove that $\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\ldots+\tan^{-1}\left(\frac{d}{1+a_na_{n-1}}\right)\right]=\frac{a_n-a_1}{1+a_1a_n}$.

Solution

Now, $\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)=\tan^{-1}\left(\frac{a_2-a_1}{1+a_la_2}\right)=\tan^{-1}a_2-\tan^{-1}a_1$.

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